

JEE Calculus, Vectors & 3D Geometry Master Compendium

Target Exam: JEE Main & JEE Advanced 2026 | Stream: Engineering | Subject: Mathematics

Verified Official Academic Preparation Material - TheCampusNova Learning Portal

Key Topics Covered: Definite Integration, Differential Equations & 3D Vectors

1. Definite Integration King Property: $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$
2. Leibniz Rule: $\frac{d}{dx} [\int_{u(x)}^{v(x)} f(t)dt] = f(v(x)) \cdot v'(x) - f(u(x)) \cdot u'(x)$
3. Vectors: Dot Product ($a \cdot b = |a||b|\cos \theta$) | Cross Product ($a \times b = |a||b|\sin \theta \cdot \hat{n}$)
4. 3D Line Equation: $\frac{(x - x_1)}{l} = \frac{(y - y_1)}{m} = \frac{(z - z_1)}{n}$ | Shortest distance between skew lines formula
5. Solved PYQ: Evaluate $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{(\sqrt{\sin x} + \sqrt{\cos x})} dx$. Answer: $\pi / 4$
6. Exam Strategy: Do not get stuck on lengthy Coordinate Geometry problems; Vectors & 3D Geometry take under 3 minutes per MCQ.